

Randomized, controlled trial promotes physical activity and reduces consumption of sweets and sodium among overweight and obese adults

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Abstract

The present study sought to assess the impact of an intervention to reduce weight and control risk factors of noncommunicable chronic diseases in overweight or obese adults who are users of primary and secondary healthcare units of the public health system of Pelotas, Brazil. We hypothesized that individuals who received an educational intervention regarding how to lose weight and prevent other noncommunicable chronic disease risk factors through nutrition would lose weight and acquire active habits during leisure time more frequently than individuals under regular care. Two hundred forty-one participants from the Nutrition Outpatient Clinic of the Medical Teaching Hospital of the Federal University of Pelotas, Brazil, aged 20 years or older and classified as overweight or obese were randomly allocated to either the intervention group (IG; $n = 120$) or control group (CG; $n = 121$). The IG received individualized nutritional care for 6 months, and the CG received individualized usual care of the health services. Intention-to-treat analyses showed that at 6 months, mean fasting glycemia and daily consumption of sweet foods and sodium were reduced, and the time spent on physical leisure activity was increased in IG. Analysis of adherence to the protocol of the study revealed that individuals from IG had lost more in body weight, waist circumference, and fasting glucose compared to the CG. Leisure time physical activity increased in IG. Individuals adhered equally to the dietetic recommendations, irrespective of the nutrition approach that was used.

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Keywords:

Nutrition; Randomized clinical trial; Self-care; Obesity; Chronic diseases

Abbreviations:

NCD, Non-communicable disease; AN-UFPel, Nutrition Outpatient Clinic of Medical Teaching Hospital of the Federal University of Pelotas–Brazil; BMI, Body mass index; IG, Intervention group; CG, Control group; PAHO, Pan-American Health Organization; BP, Blood pressure; LDL, Low-density lipoprotein.

1. Introduction

Chronic diseases represent the leading cause of death and incapacitation worldwide [1]. Multiple risk factors, including

sedentarism, excessive consumption of saturated fat and low intake of fruit and vegetables, have been linked to the development of these diseases [1,2].

Obesity is the main component in the incidence of chronic diseases, affecting practically all age and socioeconomic groups. Approximately 90% of those with type 2 diabetes mellitus are obese or overweight [3,4]. Several

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types of cancer, particularly those located in the intestine, are associated with being overweight. Dyslipidemia and arterial hypertension constitute independent risk factors for cardiovascular disease and are often found in individuals who are overweight or obese [5].

Thus, nutritional interventions aimed at preventing or treating overweight and obesity can potentially lower the occurrence of chronic diseases, especially hypertension [6], dyslipidemia and insulin resistance, by reducing the prevalence of their respective risk factors [7]. The present study sought to assess the impact of a nutritional intervention that strived to reduce weight and to control risk factors of noncommunicable chronic diseases (NCD) in overweight or obese adults, who are clients of primary and secondary healthcare units of the public health system of the city of Pelotas, in the state of Rio Grande do Sul, Brazil. The hypothesis was that individuals who received an educational intervention regarding how to lose weight and prevent other NCD risk factors through nutrition would lose weight and acquire active habits during leisure time more frequently than individuals under regular care.

2. Methods and materials

An unblind, randomized, controlled clinical trial was conducted between June 2005 and December 2006 at the Nutrition Outpatient Clinic of the Medical Teaching Hospital of the Federal University of Pelotas–Brazil (AN-UFPel), which provides health care under the national public health system. Participants were 20 years of age or older, classified as overweight or obese (body mass index [BMI] $>24.9 \text{ kg/m}^2$) and were referred by nutritionists of the public health system as individuals who had previously sought treatment to lose weight. Patients whose dietetic prescriptions required changes in recommended levels of macronutrients to treat a specific disease (such as diabetes mellitus, acquired immunodeficiency syndrome, cancer, and hepatic or renal diseases) were excluded, as were pregnant or lactating mothers.

2.1. Sample size

The sample was calculated to detect a reduction of at least 30% in the proportion of overweight and obese participants in the intervention group (IG) and 15% in the control group (CG) (given this group were also to receive individualized nutritional care), and yield a 95% confidence level and statistical power of 80%. Based on these criteria, an estimated 120 individuals would be required in each group to give a predicted 36 and 18 individuals in the IG and CG, respectively, who would present a BMI below 25 kg/m^2 after 6 months.

2.2 Sample selection and baseline measures

Nutritionists responsible for all primary and secondary units of the municipal's public network were contacted

and informed about the study. These specialists were instructed to refer patients meeting the selection criteria to the AN-UFPel.

Patients who accepted the invitation were initially assessed. Demographic characteristics included gender, marital status and age in years. The economic level was collected according to criteria by the National Association of Research Companies. This association uses ownership of assets and schooling of the head of the family to rate economic level from A to E, where A corresponds to the wealthiest class [8]. The anthropometric variables included current weight and height (measured on a Filizola brand mechanical set of scales with 150-kg capacity, fitted with an estadiometer). Waist circumference was measured in cm at midway between the lower costal margin and the iliac crest while the person was standing and at the end of a normal expiration [9]. Weekly frequency of physical leisure activity was measured using the long version of the International Physical Activity Questionnaire [10]. Weekly consumption of macro and micronutrients was assessed using a food frequency questionnaire, previously validated via the Brazilian population [11]. At the end of the consultation after the individual had at least a half hour of rest, mean blood pressure (BP) was taken via the radial artery, using a portable digital device (OMRON brand, Beijing, China; model HEM 629 Class II) according to manufacturer's instructions. Fasting sera lipid and glucose concentrations were measured in samples of venous blood and analyzed at the Laboratory of the UFPel using the automated enzymatic method (Labmax 340 device—Labtest).

Individuals whose fasting glucose was consistent with an initial diagnosis of diabetes mellitus [4] were excluded from the study and referred back to the referring clinics for medical treatment or to the reference system of the municipality. Participants who received a biochemical diagnosis of dyslipidemia [12] were advised to seek medical care to ascertain the need to run further complementary exams and/or to start drug treatment but were not excluded from the study. Before randomization, those who met the selection criteria were told that subsequent consultations would take place at the AN-UFPel (IG) or at the referring services (CG), based on random allocation.

2.3. Randomization

Randomization was carried out in blocks of 6 individuals stratified by BMI, gender and age-groups, so as to achieve a balance of these aspects for both groups. Participant allocation was defined a priori before study commencement. Each group of 6 individuals who attended the stratum characteristics outlined, had a corresponding sealed envelope which contained 6 slips; 3 bearing the status of intervention (IG) and 3 bearing the status of control (CG). After all slips had been drawn they were placed back in the envelope and reused. Individuals allocated to the IG underwent the intervention, while those allocated to the

CG were referred back to the referring service to receive the service's standard guidance based on individual consultations with the assigned nutritionist.

2.4. Intervention

The intervention was run by 2 trained nutritionists through individual consultations and involved the following 5 components: (a) dietetic prescription, (b) guidance on choosing and substituting foods, (c) encouragement to consume fruit, vegetables and low-fat foods, (d) encouragement to perform physical activity and e) promotion of adherence for follow up visits.

- a. Individualized monthly dietetic prescriptions were devised according to recommendations by the World Health Organization and Pan-American Health Organization [3,5]. Calorie controlled diets were prescribed based on expected weight for a target BMI of 21.9 kg/m² (maximum recommended value for chronic disease prevention) and on number of kJ per kilogram of body weight (125.5 kJ for men and 104.6 kJ for women) [5]. The nutrients in the diet were prescribed in order to satisfy recommended requirements, that is, 15% to 30% total fat, 55% to 75% total carbohydrates, 10% to 15% proteins, up to 300 mg/d of cholesterol, up to 5 g/d of salt (or 2000 mg/day of sodium), and up to 25 g/d of dietary fiber. Daily consumption of fruit and vegetables was determined as at least 400 g [5]. A manual was prepared especially for the study containing photographs illustrating the portion sizes of the prescribed foods. This manual was used by nutritionists as a supportive aid to assist the process of explaining the prescribed diet to patients of the IG. With the aim of verifying whether serving sizes varied according to the BMI of the individual, a substudy was performed on a convenient sample comprising 30 individuals with different BMIs who did not participate in the main study. The weight of food portions served by these individuals was compared against the weight of standard portions. The substudy identified no systematic error in measures, revealing only small positive and negative variations between served and standard measures.
- b. Participants were provided with nutritionally balanced food lists, together with calorically equivalent alternatives, in 100-cal portions. The portions of each food group were defined according to the nutritional pyramid recommended for the Brazilian population [13].
- c. Encouragement to consume fruit, vegetables and low-fat foods was given at the first consultation and was reinforced during each follow up session and consisted of explanations regarding the nutritional value of these foods [13].
- d. Encouragement to practice physical exercise consisted of recommending a minimum of 30 minutes of daily

physical leisure activity at a moderate level (preferably aerobic) to be performed at least 4 times per week. Guidance on how to introduce activities into the daily routine which raise calorific expenditure at work, in transit and during leisure time was also provided [13].

- e. Promoting adherence to follow-up sessions was achieved through prior scheduling of appointments and by telephoning absentees 15 days after missed appointments.

2.5. Follow up

Follow-up sessions were individualized in both groups, in a similar fashion to the initial consultations. The objective of the follow-up consultations in the IG was to assess adherence to the initial dietetic prescription; provide additional explanations regarding the menu and alternative foods; promote the consumption of fruit, vegetables and low-fat foods; encourage the practice of physical activity and increase adherence regarding return visits. Follow-up was carried out in the control group as the nutritionist saw fit, where regularity of consultations varied, as did conduct for each patient and the action taken in the event of absence.

2.6. The outcome

At 6 months from study baseline, the 2 groups were compared in terms of anthropometric variables; BP, sera lipids and glucose levels; physical activity; and food, calorie, and nutrient intake. All outcomes were evaluated using the same instruments and methods used at baseline assessment. Except for laboratory tests, all other outcome indicators were assessed by observers that were unblind of the status (intervention or control) of the study participants.

2.7. Compliance

Adherence to the study protocol by nutritionists was checked by observing a sample of consultations of both groups. A trained nutrition student, with the aid of a checklist based on the expected behavior of the health professionals in terms of food guidance, observed 24 consultations which were randomly chosen from each group and represented an equal number of initial and follow-up sessions.

2.8 Statistical analyses

A database was built using the Epi-info 6.0 [14] software application to store the information gathered. Food frequency questionnaires were first analyzed by the Dietsys [15] program and subsequently included in the general study database. All data were processed by means of double entry and consistency checking of information, and analyzed using the STATA program, version 10.0 [16]. Participants from the 2 groups were compared at baseline by the χ^2 , Student *t*, and Wilcoxon tests (this last test was applied when variables presented a nonnormal distribution). Intention-to-treat, adherence to protocol, and intragroup analyses were carried out (Student's *t* test or the paired Wilcoxon test were applied

in this last analysis). For continuous variables IG and CG were compared in terms of means and SEs and *t* tests were used. For categorical variables, the 2 groups were compared in terms of proportions and differences assessed through χ^2 tests. All the tests were 2 sided.

2.9. Ethics

This study was previously approved by the Research and Ethics Committee of the Federal University of Pelotas Medical School. All participants signed a written consent form before enrolling in the study. The study presented no health risk to participants. Any alterations seen on laboratory examinations were sent for further medical examination.

3. Results

The flow chart in Fig. 1 shows the different phases of the study and the participants at each stage. Three hundred twenty-seven individuals contacted AN-UFPel to take part in the study. Thirty eight did not fulfill the selection criteria for entry: 10 due to glycemia consistent with diabetes diagnosis, 24 due to BMI lower than 24.9 kg/m², 3 because they were undergoing treatment for cancer, and 1 who presented cirrhosis of the liver. Of the 289 individuals who met the study inclusion criteria, 48 (16.6%) refused to participate before the randomization stage. No difference was found between those who refused to participate and participants in terms of sex, weight, or BMI distribution. However, participants tended to be younger than those who

refused. Two hundred forty-one individuals were randomized for intervention (*n* = 120) or standard care (*n* = 121). Of these, 49 dropped out of the study during follow up, representing 19.2% (*n* = 23) in the IG and 21.5% (*n* = 26) in the CG. The reasons stated for discontinuation were lack of motivation or time to continue treatment. A total of 67 individuals from the IG attended all 6 sessions programmed. Outcome measures were performed for 97 individuals in the IG and 95 in the CG.

Table 1 shows the characteristics of participants at baseline. The 2 groups differed for fasting glucose, whose mean value was higher among IG participants, although levels were not indicative of diabetes mellitus. At baseline, 103 (85.8%) subjects in the IG and 100 (82.6%) in the CG, (*P* = .50) presented at least 1 type of dyslipidemia [11].

Table 1
Characteristics of participants at baseline, by randomization category

Variable	Intervention group (<i>n</i> = 120)	Control group (<i>n</i> = 121)	<i>P</i>
<i>Sociodemographic characteristics</i>			
Age (y)	41.1 (1.2)	39.6 (1.1)	.43
Female gender	108 (90.0)	106 (87.6)	.35
Married or with partner	80 (66.7)	79 (65.2)	.80
Economic level			.39
A	5 (4.2)	8 (6.7)	
B	61 (50.8)	53 (44.2)	
C	43 (35.8)	52 (43.3)	
D	11 (9.2)	8 (5.8)	
<i>Anthropometric and clinical characteristics</i>			
Weight (kg)	85.3 ± 1.5	87.9 (1.9)	.29
BMI (kg/m ²)	33.5 ± 0.5	34.6 (0.6)	.37
25.0–29.9	36 ± 30.0	37 ± 30.8	.91
30.0 – 39.9	66 ± 55.0	63 ± 52.5	
40 or more	18 ± 15.0	21 ± 16.7	
Waist circumference (cm)	96.9 ± 1.2	99 ± 1.3	.24
Total cholesterol (mmol/L)	5.0 ± 0.1	4.9 ± 0.1	.32
HDL cholesterol (mmol/L)	1.4 ± 0.0	1.3 ± 0.0	.23
LDL cholesterol (mmol/L)	2.9 ± 0.1	2.8 ± 0.1	.16
VLDL cholesterol (mmol/L)	0.7 ± 0.0	0.8 ± 0.0	.25
Triglycerides (mmol/L)	1.7 ± 0.1	1.8 ± 0.1	.33
Fasting glucose (mmol/L)	5.3 ± 1.1	5.1 ± 1.1	.02
Systolic BP (mmHg)	117.1 ± 1.5	115.4 ± 1.6	.38
Diastolic BP (mmHg)	73.6 ± 1.0	73.1 ± 1.1	.73
Physical leisure activity (min/wk)	67.4 ± 13.5	107.0 ± 20.2	.26
Previous history of slimming diet (yes)	83 ± 69.2	84 ± 69.5	.86
<i>Energy, nutrient and food intake</i>			
Total calories (kJ)	7055.9 ± 265.5	7150.9 ± 247.7	.79
Proteins (%)	17.0 ± 0.3	17.8 ± 0.3	.10
Fats (%)	33.8 ± 0.5	34.1 ± 0.5	.69
Carbohydrates (%)	48.7 ± 0.6	47.5 ± 0.6	.17
Cholesterol (mg)	192.5 ± 9.0	205.7 ± 8.1	.09
Fiber (g)	12.9 ± 0.6	12.5 ± 0.6	.60
Sodium (mg)	2085.6 ± 91.2	2084.7 ± 102.7	.47
Fruit (g)	183.5 ± 20.0	161.8 ± 15.4	.40
Vegetables (g)	35.4 ± 2.5	33.8 ± 2.5	.55
Sweet foods (g)	96.2 ± 12.1	96.2 ± 11.2	.78

HDL, high-density lipoprotein. Values are expressed as *n* (%) or means ± SE. *P* value: Student *t* test, Wilcoxon test, or χ^2 test.

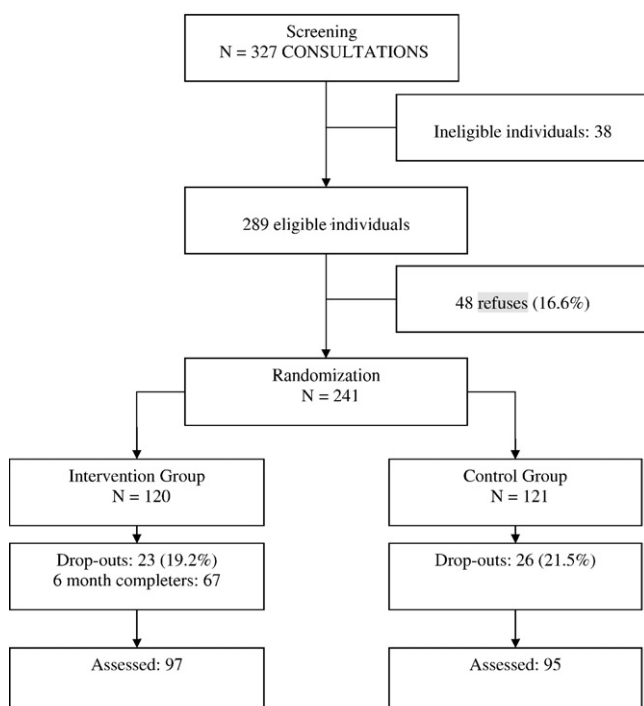


Fig. 1. Study flowchart.

Table 2

Performance of nutritionists based on consultation observations, by randomization group

Actions of nutritionists	Intervention group (n = 24)	Control group (n = 24)	P
<i>Categorical variables</i>			
Encourages patient to talk	24 (100.0)	22 (92.0)	.15
Praises beneficial acts	20 (83.3)	14 (58.3)	.06
Uses accessible language	24 (100.0)	24 (100.0)	1.00
Outlines inappropriate eating practices	24 (100.0)	20 (83.3)	.04
Explains list of substitutions	12 (100.0)	5 (41.7)	.007
Explains substitutions for each food group	12 (100.0)	6 (50.0)	.02
Explains healthy use of oils and fats ^a	5 (41.7)	5 (41.7)	1.00
Shows photos or examples of portion sizes	14 (58.3)	0 (0.00)	<.001
Provides folder or printed material containing nutritional guidance	12 (100.0)	9 (75.0)	.13
Explains material provided ^b	12 (100.0)	4 (19.0)	<.001
Emphasizes guidance at end of consultation	12 (100.0)	5 (41.7)	.03
Encourages practice of physical activity	24 (100.0)	19 (79.2)	.02
<i>Continuous variables</i>			
Length of consultation (min)	23.9 ± 1.9	22.1 ± 3.1	.29
Length of first consultation (min)	27.8 ± 2.0	32.4 ± 4.0	.47
Length of follow-up visits (min)	19.1 ± 3.0	11.8 ± 2.1	.03

VLDL, very low-density lipoprotein. Values are expressed as n (%) or means ± SE.

P value: Student *t* test, Wilcoxon test, or χ^2 test.

^a Only observation of first consultation n = 12.

^b Only consultations at which material was offered n = 21.

Of these, 12 had undergone treatment with drugs (5 in the IG and 7 in the CG).

Table 2 shows that nutritionist performance differed on most items assessed at the 24 observed consultations in each group. Higher performance was observed in the IG group. No difference was found between the 2 groups regarding length of initial consultation (23.9 vs 22.1 minutes for IG and CG, respectively).

After 6 months of the study (Table 3), the intention-to-treat analysis showed that the IG had lost an average of 2.4 kg and the CG, 2.0 kg. The mean proportion of weight lost was similar for the 2 groups: 2.5% and 2.0% for the IG and CG, respectively. Mean fasting glycemia was found to have dropped (−3.4 mmol/L) in IG individuals yet increased (+4.4 mmol/L) ($P = .001$) among individuals from the CG. At the end of the 6-month study period, 3 individuals in the IG (3.1%) were no longer overweight (BMI ≤ 24.9 kg/m²). In contrast, none of the individuals in the CG attained normal weight status.

Time spent on physical leisure activity increased by almost 1 1/2 hours in the IG, yet decreased by an average of 40 minutes in the CG ($P = .01$). Compared to the CG, there was a marked reduction among IG individuals in daily consumption of sweet foods (−44 g versus −11 g, for the IG and CG, respectively; $P = .01$) and in sodium (−371 mg/day versus +33 mg/day; $P = .03$).

Table 3

Differences in anthropometric, clinical, laboratorial and behavioral parameters between baseline and 6 months after study enrolment, by randomization category (intention to treat analysis)

Variable ^a	Intervention group (n = 97)	Control group (n = 95)	P value
<i>Anthropometric and clinical characteristics</i>			
Weight (kg)	−2.4 ± 0.5	−2.0 ± 0.4	.48
Weight loss (%)	−2.5 ± 0.5	−2.0 ± 0.5	.34
BMI (kg/m ²)	−0.9 ± 0.2	−0.8 ± 0.2	.49
Waist circumference (cm)	−4.8 ± 0.5	−3.5 ± 0.5	.06
Total cholesterol (mmol/L)	−0.0 ± 0.1	0.1 ± 0.1	.54
HDL cholesterol (mmol/L)	0.0 ± 0.0	−0.1 ± 0.0	.07
LDL cholesterol (mmol/L)	0.0 ± 0.1	0.0 ± 0.1	.85
VLDL cholesterol (mmol/L)	0.0 ± 0.0	0.1 ± 0.1	.06
Triglycerides (mmol/L)	−0.2 ± 0.1	0.0 ± 0.1	.20
Fasting glucose (mmol/L)	−3.4 ± 1.2	4.4 ± 3.1	.01
<i>BP</i>			
Systolic (mm Hg)	−3.4 ± 1.6	−4.9 ± 1.0	.63
Diastolic (mm Hg)	−3.8 ± 1.1	−4.9 ± 1.0	.47
Physical leisure activity (min/wk)	88.4 ± 35.2	−38.7 ± 27.6	.01
<i>Energy, nutrient and food intake</i>			
Total kJ	−1299.1 ± 301.2	−421.7 ± 328.0	.15
Proteins (%)	2.0 ± 0.5	0.9 ± 0.6	.14
Fats (%)	−2.5 ± 0.8	−2.6 ± 0.8	.95
Carbohydrates (%)	0.2 ± 0.9	1.7 ± 1.0	.26
Cholesterol (mg)	−29.3 ± 9.5	−29.2 ± 10.4	.84
Fiber (g)	−1.1 ± 0.7	0.6 ± 0.8	.12
Sodium (mg)	−371.5 ± 112.7	33.0 ± 109.1	.03
Fruit (g)	−52.0 ± 19.6	3.7 ± 28.5	.15
Vegetables (g)	12.1 ± 3.8	11.1 ± 5.0	.76
Sweet foods (g)	−44.2 ± 12.7	−11.0 ± 19.2	.01

Values are expressed as means ± SE.

P value: Student *t* test, Wilcoxon test, or χ^2 test.

^a Differences between end and initial value.

Comparison of subjects from the IG who attended all consultations (n = 67), against CG subjects (analysis of adherence to the protocol; data not shown in table) revealed statistically significant improvements in the intervention group for outcomes of percentage weight loss (−3.5 in the IG vs −2.0 in the CG; $P = .05$), waist circumference (−6.0 cm in the IG versus −3.5 cm in the CG; $P = .004$), fasting glucose (−3.3 mmol/L in the IG versus +4.4 mmol/L in the CG; $P = .02$) and physical activity (+127.8 minutes/week versus −38.7 minutes per week in the CG; $P = .001$). Akin to the intention-to-treat analysis, positive improvements in the other parameters were observed in the IG, although these did not reach statistical significance. The food intake analysis found the same results as the intention-to-treat analysis.

The intragroup analyses presented in Table 4 shows that both groups had statistically significant reductions in cholesterol intake and an increased level of vegetable consumption. A noteworthy observation is the low daily fruit and vegetable intake among individuals of both groups, which fell far below the currently recommended levels [5]. Other relevant factors were the higher-than-recommended

Table 4

Anthropometric, clinical, laboratorial and behavioral parameters at study baseline (T1) and endpoint (T2) (6 months later), stratified by randomization category (intragroup analysis)

Variable	Intervention group (n = 97)			Control group (n = 95)		
	T1	T6	P value	T1	T6	P
<i>Anthropometric and clinical characteristics</i>						
Weight (kg)	85.0 ± 1.7	82.6 ± 1.7	<.001	86.2 ± 1.9	84.3 ± 1.8	<.001
BMI (kg/m ²)	33.6 ± 0.6	32.7 ± 0.6	<.001	33.9 ± 0.6	33.2 ± 0.6	<.001
Waist circumference (cm)	97.0 ± 1.4	92.1 ± 1.2	<.001	98.3 ± 1.4	94.8 ± 1.4	<.001
Total cholesterol (mmol/L)	5.5 ± 0.1	5.1 ± 0.1	.90	4.9 ± 0.1	5.0 ± 0.1	.51
HDL cholesterol (mmol/L)	1.4 ± 0.0	1.4 ± 0.0	.32	1.3 ± 0.0	1.3 ± 0.0	.11
LDL cholesterol (mmol/L)	3.0 ± 0.1	3.0 ± 0.1	.47	2.8 ± 0.1	2.8 ± 0.1	.82
VLDL cholesterol (mmol/L)	0.7 ± 0.0	0.6 ± 0.0	.22	0.8 ± 0.0	0.8 ± 0.1	.66
Triglycerides (mmol/L)	1.7 ± 0.1	3.4 ± 0.1	.05	1.8 ± 0.1	1.8 ± 0.1	.77
Fasting glucose (mmol/L)	5.3 ± 1.3	5.1 ± (1.1)	.01	5.1 ± 1.2	5.3 ± 3.0	.16
Systolic Blood Pressure (mm Hg)	117.5 ± 1.8	114.1 ± 1.7	.01	115.9 ± 1.7	111.5 ± 1.6	.20
Diastolic Blood Pressure (mm Hg)	74.0 ± 1.2	70.2 ± 1.2	<.001	73.2 ± 1.2	68.3 ± 1.1	<.001
<i>Physical leisure activity (min/week)</i>						
	69.5 ± 15.0	157.8 ± 34.2	.002	130.8 ± 24.1	92.1 ± 18.0	.54
<i>Energy, nutrient, and food intake</i>						
Calories (kJ)	6963.0 ± 308.8	5663.9 ± 238.1	<.001	6372.2 ± 307.1	5950.5 ± 20.1	.16
Proteins (%)	17.3 ± 0.4	19.3 ± 0.5	<.001	17.6 ± 0.4	18.5 ± 0.5	.14
Fats (%)	31.3 ± 0.7	33.8 ± 0.6	.002	30.7 ± 0.6	33.3 ± 0.7	.002
Carbohydrates (%)	48.6 ± 0.7	48.8 ± 0.8	.85	48.6 ± 0.8	50.3 ± 0.8	.10
Cholesterol (mg)	190.5 ± 10.0	161.2 ± 8.5	.002	181.0 ± 10.4	151.8 ± 8.5	.007
Fiber (g)	12.8 ± 0.7	11.7 ± 0.5	.32	12.1 ± 0.7	12.7 ± 0.8	.39
Sodium (mg)	2111.2 ± 117.0	1739.6 ± 89.3	.002	1681.3 ± 96.8	1714.3 ± 202.3	.84
Fruit (g)	186. ± 20.1	133.9 ± 11.2	.02	155.7 ± 21.6	159.4 ± 20.5	.90
Vegetables (g)	34.3 ± 3.0	46.4 ± 3.3	.003	34.4 ± 4.0	45.5 ± 4.4	.02
Sweet foods (g)	81.5 ± 12.9	37.3 ± 5.4	<.001	101.5 ± 23.3	90.5 ± 19.9	.79

Values are expressed as n (%) or means ± SE. P value: Student *t* test, Wilcoxon test, or χ^2 test.

percentages of proteins and lipids, and lower-than-recommended levels of carbohydrates [5] which remained unchanged after 6 months. Cholesterol consumption did not exceed recommended maximum limits in either of the 2 groups. Both groups presented statistically significant reductions in weight, BMI, waist circumference and diastolic BP. A slight increase in percentage of fat intake was observed in both groups. Overall, the IG presented better results than the CG group, after 6 months of follow up.

4. Discussion

This randomized clinical trial refuted the hypothesis of greater weight loss but supported the hypothesis that an educational intervention may promote the adoption of healthy behaviors in comparison with standard dietetic prescription provided by nutritionists. The reasons for this lack of improvement shall be interpreted under the light of the methodology used in the study and the potential efficacy of the intervention in comparison to the care typically provided by nutritionists under the public health system.

Concerning the methodological aspects, issues like the comparability of the groups at baseline, the randomization process, adherence to the dietetic prescription both by nutritionists and participants, length of follow-up and the

statistical power of the study are all factors to take into account. In the present study, the randomization process in blocks and stratified by known risk factors for overweight (age, sex, and BMI), allied with the rigorous use of selection factors, ensured the comparability of the groups before commencing the intervention.

The nutritionists involved in the IG demonstrated adherence to the study protocol as evident from observation of the consultations. Conduct such as praising beneficial eating practices and discussing inappropriate practices were more frequently employed during consultations given by nutritionists for the IG. Specific recommendations on diet and ways of better understanding the diet were also more often addressed by nutritionists from the IG than the CG.

Despite the strategies used in the IG to improve adherence of individuals and to facilitate understanding of the dietetic guidance given, losses during follow-up were similar in both groups and in-line with drop-out figures reported by other study [17]. After the 6-month study period, adherence to dietetic prescriptions (verified using a weekly food consumption questionnaire) was also found to be similar for both groups, except for lower salt and sweet foods consumption reported by the IG. With the exception of fiber consumption, which remained unchanged, and fruit intake which fell during the 6-month follow up, the intragroup analyses showed that the IG adhered well to

the dietetic recommendations. Nevertheless, the main aim of this study, that is, the promotion of weight loss, was not achieved.

Among the limitations of the current study is the duration of follow-up. Six months may be too short a time for overweight or obese individuals free to self-manage their diet to lose a greater amount of body weight than subjects submitted to conventional dietetic treatment approaches. Nevertheless, studies suggest that irrespective of intervention type (behavioral, dietetic or pharmacologic); maximum weight loss tends to be observed within a 6-month timeframe [18,19]. The number of individuals that were followed up for 6 months and fell below the threshold stipulated by the sample size calculation may have reduced the statistical power of the study. However, post hoc analyses showed that for most of the outcomes investigated, the trial had 90% or greater power to detect statistically significant differences between intervention and control groups. The only exceptions were total cholesterol and consumption of fruits (respectively, 50% and 34% power).

Some authors have described favorable results from dietetic counseling for weight reduction. However, successful clinical trials have, in general, compared one group that received individualized intervention (IG) with another which received only generic, non-individualized, non-systematic nutritional guidance on weight loss strategies. Wolf et al [20] conducted a controlled, randomized clinical trial which assessed 147 individuals with type 2 diabetes and BMI 27 kg/m^2 or greater, who were members of a private health plan. These individuals were divided into 2 groups, with the IG receiving a total of 4 hours' individual care from a nutritionist, spread over 6 sessions throughout the year, in addition to 6 meetings in small groups over the same period. In the study, brief telephone calls were made on a monthly basis to the IG subjects in order to provide additional support and guidance. The CG received the standard care offered by the Health System of the University of Virginia (USA) which included educational materials while patients were free to take part in other weight loss or care programs for diabetics. Among the several findings of the cited study were the primary outcomes of reduced weight and waist circumference, which were significantly greater in IG subjects at 12 months' follow up (weight: -2.4 kg in IG versus -0.6 kg in CG, and waist circumference: -5.5 cm in IG versus -1.4 cm in CG, both with $P < .001$).

In Brazil, a randomized clinical trial conducted in a primary care center investigated 104 non-diabetic adults who were first degree relatives of individuals with diabetes or who were overweight or obese (BMI $24\text{--}35 \text{ kg/m}^2$). At the start of the study, all participants received information both in the form of written material and a 30-minute group session on leading a healthy life style. After randomization, the IG received 3 nutritional counseling sessions administered by a nutritionist during the first 6 months of follow up. The CG participants received no other type of intervention until study end-point. After 6 months, greater reduction in

body weight, waist circumference, diastolic BP, fasting glycemia, total cholesterol, low-density lipoprotein (LDL) cholesterol as well as energy, total and saturated fat intake was observed in the IG subjects [21].

Martinez and collaborators [22] studied 140 individuals in Chile who had undergone myocardial revascularization. These patients were randomized into 2 groups: the IG received 5 individualized nutritional guidance sessions given by a nutritionist, according to the recommendations of the National Cholesterol Education Program [12]; whereas, the CG was seen by a physician and provided with food tips contained in a folder published by a pharmaceutical company. No changes in sera triglyceride levels, weight or BMI were evident after 1 year in both groups. However, reduced LDL cholesterol levels were observed but only among participants of the IG.

In another trial conducted with obese women in Brazil, in which both groups received nutritional intervention and psychological support, differing only in terms of a physical exercise program offered only to IG participants [23]. 35% of the women who practiced exercise for 20 weeks lost more than 5% of their body weight, while only 10% of women from the CG achieved this same level of reduction, a result which did not reach statistical significance ($P = .06$).

In the present study, both groups received individualized care administered by nutritionists, which may explain the lack of statistically significant differences between intervention and control groups regarding reduction in overweight and obese participants. The intervention was based on recommendations that have been proven to be effective for the prevention of chronic diseases [24,25] and that are widely employed by health professionals. The 2 approaches adopted in our study differed radically only in terms of the systematic nature of the dietetic counseling employed and the resources used, which sought to promote motivation, participation and self-care [26] in a bid to improve adherence.

Corroborating with our findings, a study devised to measure the effect of motivational interviews on promoting adherence regarding dietetic counseling carried out in American women of African origin showed disappointing results [17]. Participants in the study were randomized to receive 4 individual sessions of motivational interviews or health education, in addition to a 16-session weight-loss program. Although adherence to the motivational interview and health education sessions was the same for both groups, changes in diet composition, physical activity and weight differed between the 2 groups.

The minimum body weight loss of 5% or more, recognized as significant for reducing the occurrence of chronic diseases [27], was similar between groups, having been reached by 28 (28.9%) IG individuals and 21 (22.8%) GC subjects ($P = .90$). It is possible that the intervention was insufficient to improve upon the already high quality of standard dietetic counseling currently provided under the city's public health system. In addition, the management of

weight possibly calls for alternative approaches to those used in the current study. Perhaps it is necessary to deploy and integrate models of cognitive social theory and professional training in order to acquire the technical skills needed for motivating individuals to make the desired changes in eating behavior [28]. In the current study, all the participants of both groups sought nutritional treatment voluntarily. Although this indicates their desire to lose weight, it does not necessarily reflect their wish to change nutritional habits. However, adherence to practicing physical activity, which also required behavioral changes and was not a primary outcome of the study, was substantially higher in the IG than the CG. There is a general consensus that practice of physical activity alone does not promote weight loss, but plays a role in maintaining any loss [1,19,22] and may be regarded as a beneficial effect of the intervention. The significant reduction in sodium and sweet food consumption observed among individuals from the IG is also an example of behavioral change and another beneficial aspect of the intervention. Sodium-rich diets are linked to a high incidence of arterial hypertension and clinical decompensation in individuals with high BP [2,5].

Although selection bias may have played a role in some of the observed results, analysis of adherence to the protocol showed a range of positive outcomes among individuals from the IG. Moreover, intra-group analyses found a higher number of positive outcomes among IG individuals.

In summary, this study showed that individuals adhered equally to the dietetic recommendations, irrespective of which nutrition approach used. Therefore, the management of obesity shall continue to represent a challenge to overcome in the endeavor to reduce the burden of NCD in our milieu.

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